



BCS2163 OBJECT ORIENTED DESIGN & IMPLEMENTATION

FACULTY OF COMPUTING

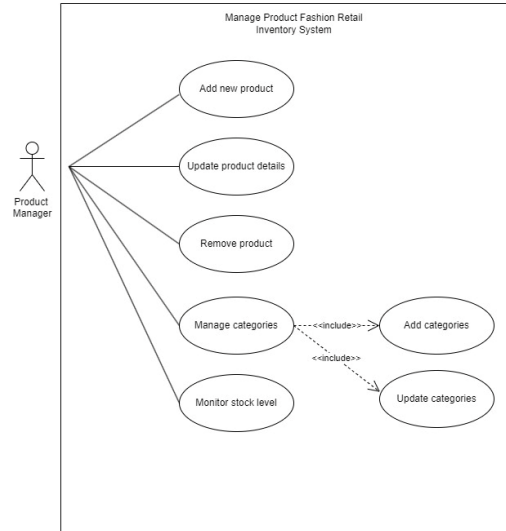
LECTURER'S NAME: DR ROZLINA BINTI MOHAMED

SECTION: 1A

	NAME	NO MATRIC
1	MARSHA BINTI FAUDZI	CD25075
2	MUHAMMAD NIZAM BIN NORDDIN	CF25008
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1.0 PRODUCT MANAGEMENT

1.1 USE CASE DIAGRAM



1.2 USE CASE DESCRIPTION

FIELD	DESCRIPTION
Use case name	Add new product
Brief description	This use case is to add new product into inventory system with details such as name, price, size and stock level
Actor	Product manager
Pre-condition	The product manager must be successfully logged into the system
Basic flow	1. Manager selects the “Add Product” option. 2. System displays a form for product code and product details. 3. Product manager enters the product name, price, category and quantity. 4. Product manager clicks the “Save” button. 5. System validates the input and save the record. 6. System display confirm successful message.

Alternate flow	A1: Cancel addition 1. Product manager clicks the “Cancel” button before saving. 2. System will return to main dashboard without saving any data.
Exception flow	E1: Duplicate product code 1. System detects the product code already exists. 2. System displays an error message “product code already exists”. 3. System prompts to enter the new product code.
Post condition	The new product record is saved in the database and is visible in the product list.
Constraints	Product price must be a positive numeric value and product code must be unique.

FIELD	DESCRIPTION
Use case name	Update product details
Brief description	This use case is to modify existing product’s attributes (price, stock, size, color, etc)
Actor	Product manager
Pre-condition	Product exists in the system
Basic flow	1. Manager search product. 2. System displays product lists. 3. Product manager selects a product. 4. System shows editable form. 5. Manager modifies fields. 6. Product manager saves the product details. 7. System validates and updates.
Alternate flow	A1: Cancel update 1. Manager clicks "Cancel" instead of save. 2. System reverts changes and returns to list.
Exception flow	E1: Wrong price value 1. System detects negative price 2. System will revert changes and notify error
Post condition	Product already existed in system
Constraints	Only manager can update details

FIELD	DESCRIPTION
Use case name	Remove product
Brief description	This use case is to delete a product from the system such as discontinued fashion item.
Actor	Product manager
Pre-condition	The product must exist and not part of any pending sales order.
Basic flow	1. Product manager searches product. 2. System shows product. 3. Product manager clicks "Remove" button. 4. System asks for confirmation. 5. Manager confirms. 6. System automatically removes product.
Alternate flow	A1: Cancel removal 1. Manager clicks "No" button at confirmation. 2. System keeps the product record.
Exception flow	None
Post condition	The product is marked inactive or deleted

FIELD	DESCRIPTION
Use case name	Manage categories
Brief description	This use case allows the manager to organize fashion items into groups like "Summer Wear".
Actor	Product manager
Pre-condition	The system is active and the product manager has access to the category dashboard
Basic flow	1. Manager selects manage categories. 2. System displays existing categories. 3. Manager choose to add and update a category. 4. System processes the specific action. 5. Manager confirms the changes. 6. Categories list updated.
Alternate flow	A1: View only 1. Manager views the categories and exits without making changes.
Exception flow	E1: Update category in use

	1. Manager attempts to update the existing category that still contains active products 2. System will change the category and prompts the manager to reassign product first.
Post condition	The category list is updated and reflected in the product.
Constraints	Category names cannot be blank and must be unique.

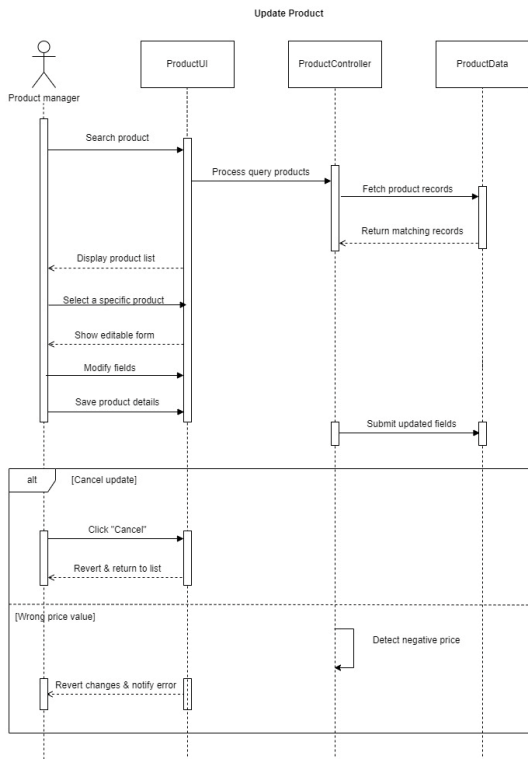
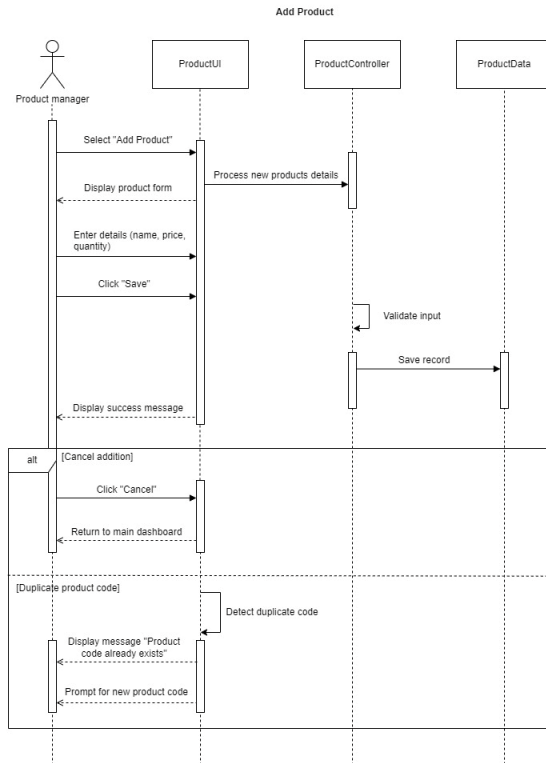
FIELD	DESCRIPTION
Use case name	Add categories
Brief description	Creates a new classification for products in the system.
Actor	Product manager
Pre-condition	System is in category management mode.
Basic flow	1. Manager clicks "Add New Category". 2. System displays name input field. 3. Manager enters category name. 4. Manager clicks "Confirm" button. 5. System validates and adds category to list.
Alternate flow	A1: Discard new category 1. Manager leaves name blank and clicks "Back" button. 2. System returns to dashboard.
Exception flow	E1: Duplicate category name 1. System detects name already exists; displays "Name Existed" error.
Post condition	New category is available for product assignment.
Constraints	Maximum category name length is 50 characters.

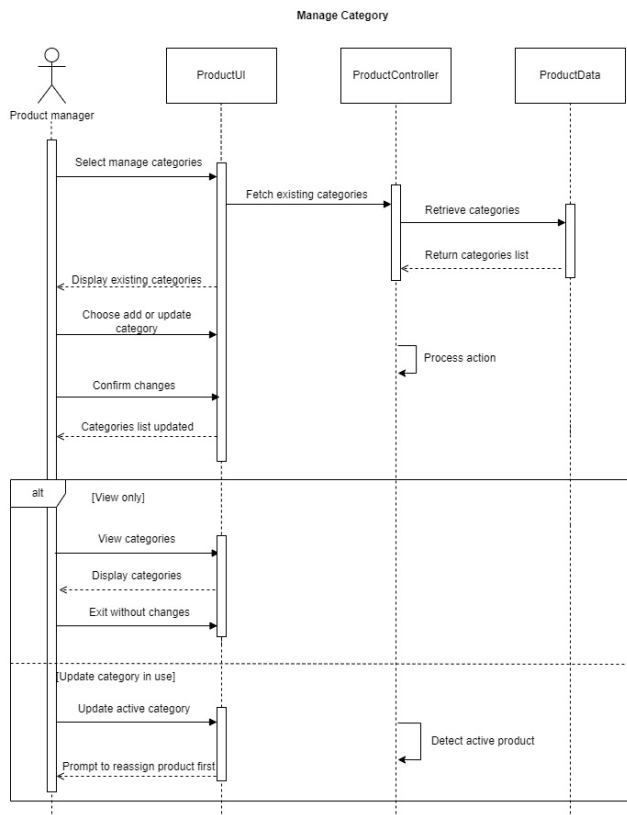
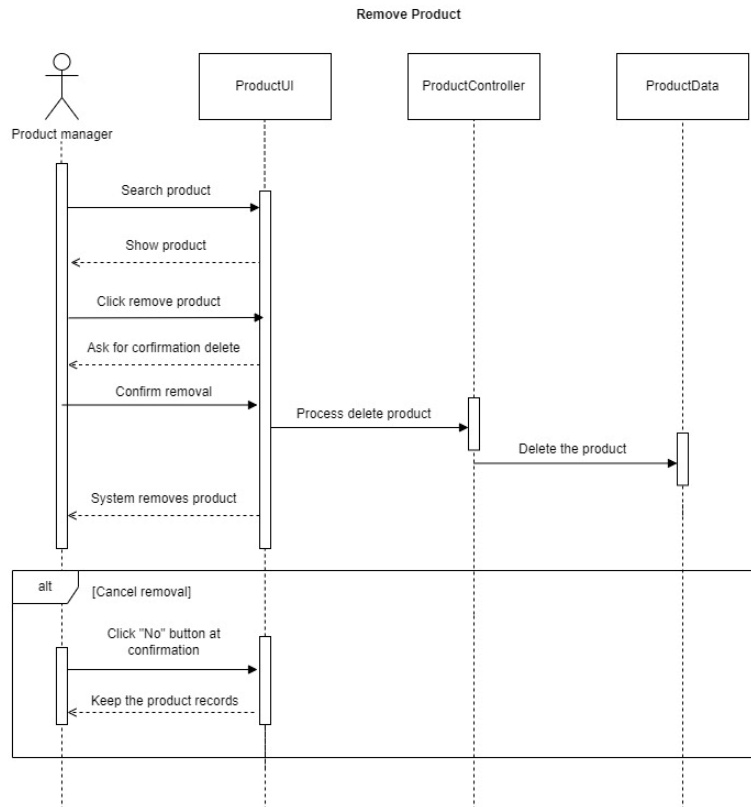
FIELD	DESCRIPTION
Use case name	Update categories
Brief description	Renames an existing category.
Actor	Product manager
Pre-condition	Category already exists in the system.
Basic flow	1. Manager selects an existing category. 2. Manager enters a new name. 3. Manager clicks "Update".

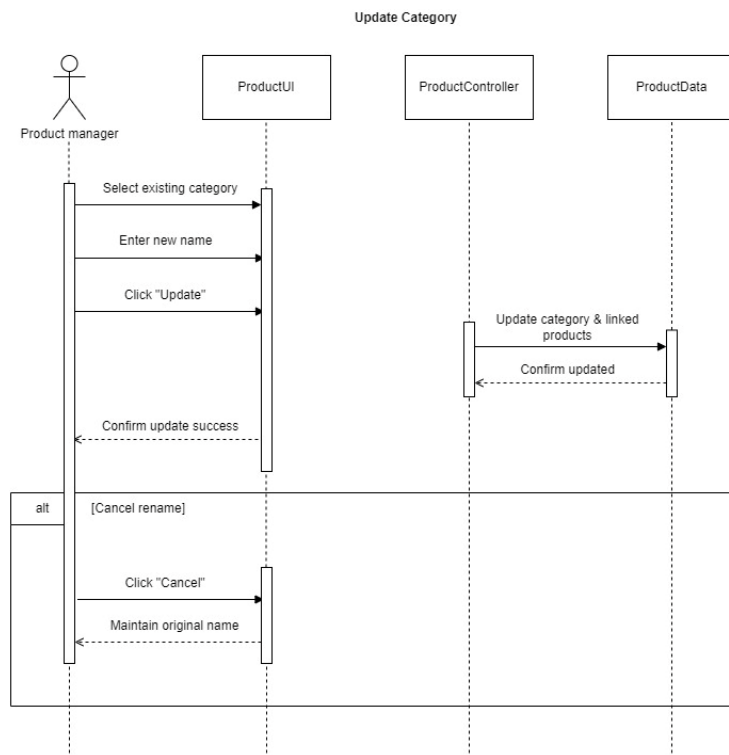
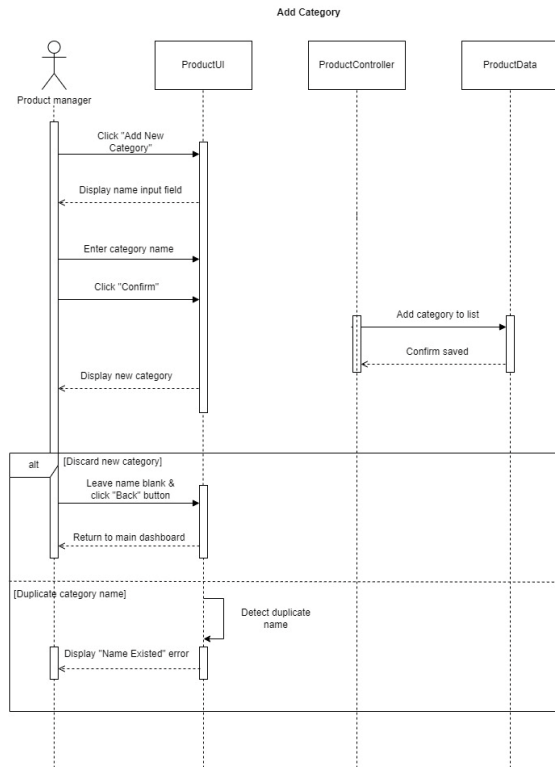
	4. System updates all linked products with the new name. 5. System confirms update.
Alternate flow	A1: Cancel rename 1. Manager selects "Cancel" button. 2. The original name is maintained.
Exception flow	None
Post condition	Category name is updated throughout the system.
Constraints	Modification restricted to admin-level managers.

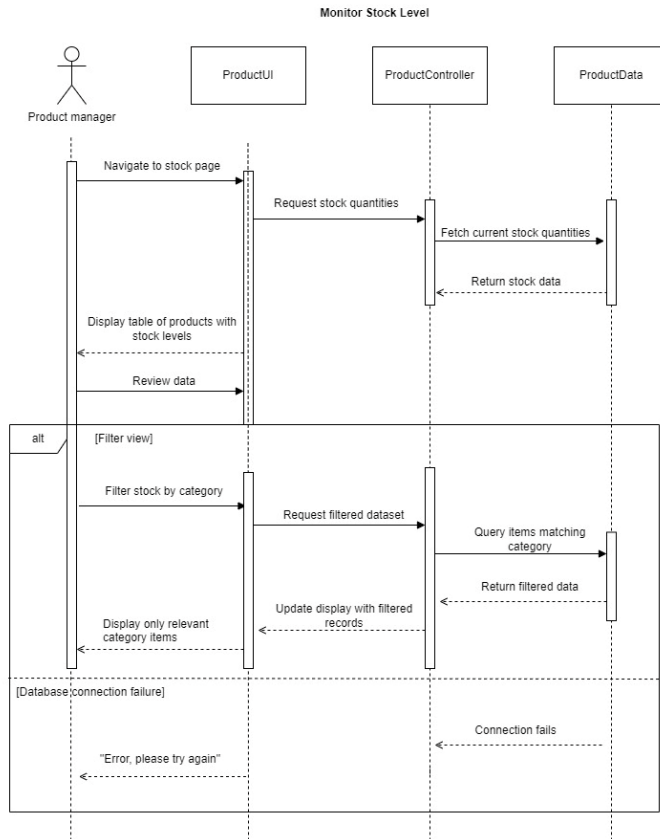
FIELD	DESCRIPTION
Use case name	Monitor stock level
Brief description	This use case allows the manager to view current inventory counts for all products to ensure stock availability.
Actor	Product manager
Pre-condition	The product record existed in database.
Basic flow	1. Manager navigates to the "Stock" page. 2. System retrieves current stock quantities for all product code. 3. System displays a table of products with their corresponding stock levels. 4. Manager reviews the data.
Alternate flow	A1: Filter view 1. Manager filters stock by category like "Sweater" and system displays only relevant items.
Exception flow	E1: Database connection failure 1. System fails to retrieve stock data. 2. System displays "Error, please try again" to show unable to fetch inventory data.
Post condition	The manager is informed of current stock status.
Constraints	The display must refresh in real-time or manual refresh request

1.3 SEQUENCE DIAGRAM









2.0 SUPPLIER MANAGEMENT

2.1 USE CASE DIAGRAM



2.2 USE CASE DESCRIPTION

Field	Description
Use Case Name	Register New Supplier
Primary Actor	Manager
Secondary Actor	None
Brief Description	Allows the manager to onboard a new supplier by saving their official company credentials into the central system database.
Pre-condition	The Manager is authenticated and has administrative profiling privileges active.
Post-condition	A new supplier profile record is initialized with a status tag of Active.
Basic Flow	1. Manager selects the "Register Supplier" option from the management control panel. 2. System renders a clean registration data entry layout. 3. Manager inputs company data strings (Company Name,

	Registration Number, Contact details). 4. Manager clicks the "Save Profile" action control. 5. System authenticates data types, archives records, and generates a unique Supplier_ID.
Alternate Flow	None
Exception Flow	Duplicate Registration Number: 1. System finds that the registration number already exists. 2. System shows an error message: "This supplier is already registered." 3. The process stops.
Constraints	System must run real-time background validation checks to prevent duplicate Company Registration Numbers.

Field	Description
Use Case Name	Manage Supplier Profile
Primary Actor	Manager
Secondary Actor	None
Brief Description	Enables the manager to update contact numbers, bank profiles, or terminate/archive inactive supplier records.
Pre-condition	The targeted supplier data profile record already exists within the active system environment database layers.
Post-condition	The targeted database row records reflect updated transactional parameters or structural modifications.
Basic Flow	1. Manager loads the master supplier record directory layout. 2. Manager picks a specific supplier row and clicks "Modify Profile". 3. System loads existing field string data elements onto input text areas. 4. Manager applies administrative updates to fields (e.g., changes telephone string details). 5. Manager commits changes by selecting "Update Changes". 6. System saves revisions and displays a modification log notification update notice.
Alternate Flow	Deactivate Supplier: 1. Manager chooses to deactivate instead of editing details.

	2. System changes supplier status to "Inactive". 3. System goes back to the supplier list.
Exception Flow	None
Constraints	Active profiles linked to historical open purchase orders cannot be permanently purged or deleted from the live system.

Field	Description
Use Case Name	Create Purchase Order
Primary Actor	Manager
Secondary Actor	Supplier
Brief Description	Orchestrates the primary structural workflow to compile item purchase lists and automatically dispatch structured order files to vendors.
Pre-condition	The supplier and items must already exist in the database.
Post-condition	The purchase order is created, saved, and sent to the supplier. Status becomes "Pending".
Basic Flow	1. Manager opens the purchase order creation page. 2. System runs the Verify Stock Reorder Level function. 3. Manager selects a supplier from the drop-down list. 4. Manager selects the items and enters the quantity to order. 5. System automatically calculates and shows the total price. 6. Manager clicks the "Submit Order" button. 7. System saves the order, creates a unique PO number, and sends a notification to the Supplier .
Alternate Flow	Save as Draft: 1. Manager clicks "Save Draft" instead of submitting. 2. System saves the data as a draft without sending it to the supplier.
Exception Flow	Invalid Quantity: 1. Manager enters a negative number or zero for quantity. 2. System blocks submission and displays: "Quantity must be more than 0."
Constraints	The order cannot be submitted if any mandatory field is empty.

Field	Description
Use Case Name	Verify Stock Reorder Level
Primary Actor	Manager
Secondary Actor	None
Brief Description	System automatically checks the inventory level to confirm if the items are low in stock and need to be ordered.
Pre-condition	Triggered automatically when starting the Create Purchase Order process.
Post-condition	System displays the items that are low in stock.
Basic Flow	1. System checks the current stock level in the inventory database. 2. System shows a warning list of items that are below the safety level. 3. System adds these items to the purchase order form layout.
Alternate Flow	None
Exception Flow	Database Connection Offline: 1. System fails to connect to the inventory database. 2. System shows an error: "Database offline. Cannot check stock level." 3. The process stops.
Constraints	Stock quantity values must always be displayed as positive numbers.

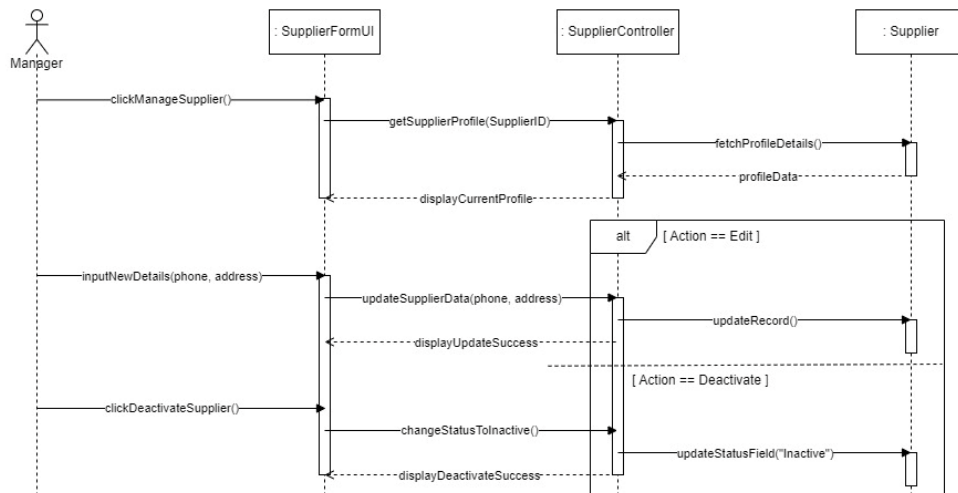
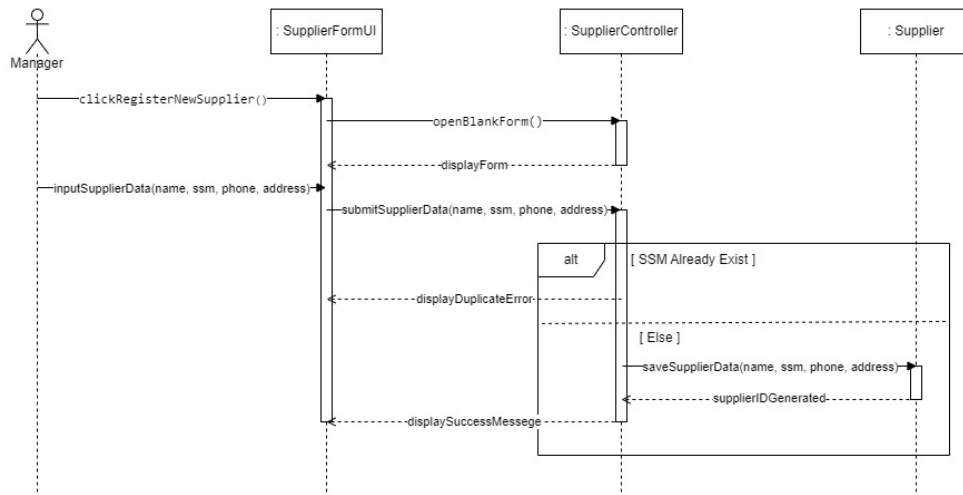
Field	Description
Use Case Name	Approve High Value PO
Primary Actor	Manager
Secondary Actor	None
Brief Description	Adds a special approval step if the total purchase order cost is higher than normal corporate budget limits.
Pre-condition	The total cost calculated in Create Purchase Order is more than RM5,000.
Post-condition	The purchase order is approved and released to be sent to the supplier.
Basic Flow	1. System detects the total cost is more than RM5,000. 2. System pops up an approval window asking for authorization. 3. Manager enters their manager password/PIN and clicks "Approve". 4. System unlocks the order and continues with the normal PO creation.
Alternate Flow	Order Rejected: 1. Manager decides to click "Reject" because it exceeds the budget. 2. System updates the status to "Rejected" and cancels the order.
Exception Flow	Wrong Password: 1. Manager enters the wrong authorization password. 2. System shows an error message: "Incorrect password. Approval denied."
Constraints	Only a user logged in with a Manager account can authorize high-value orders.

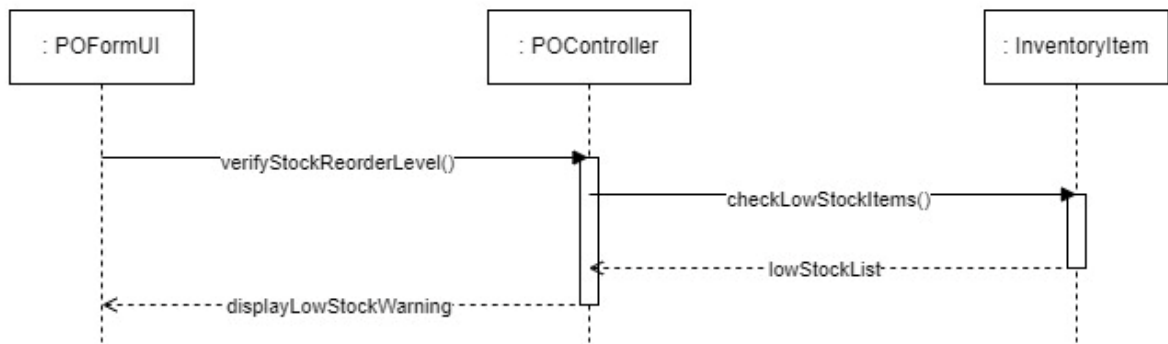
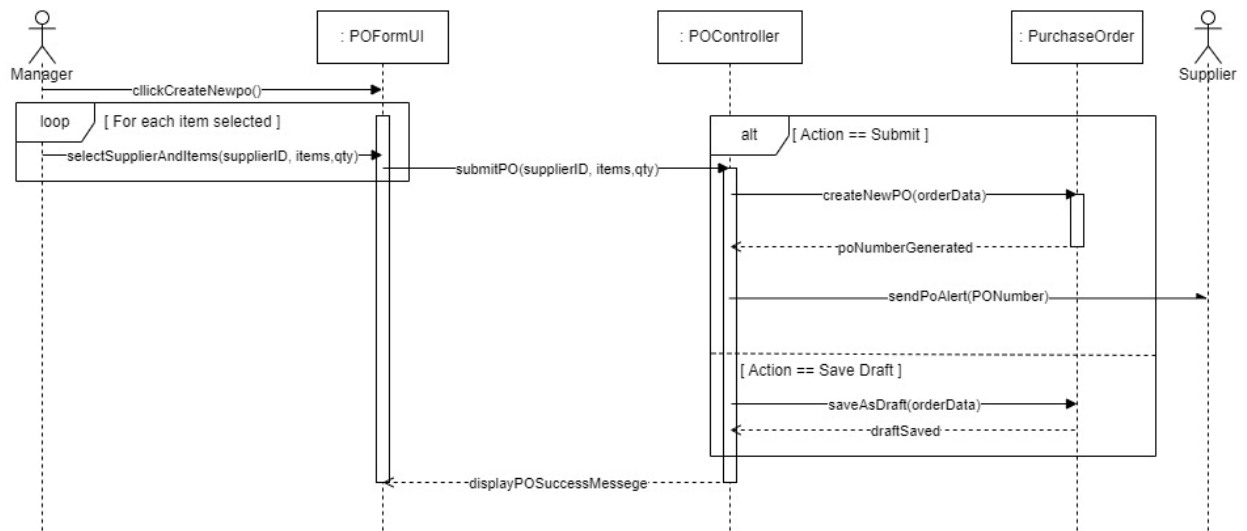
Field	Description
Use Case Name	Track PO Status
Primary Actor	Manager
Secondary Actor	Supplier
Brief Description	Allows both the manager and supplier to view the current status of a purchase order (e.g., Pending, Shipped, Delivered).
Pre-condition	The purchase order already exists in the system.
Post-condition	The current order status tracking timeline is shown on the screen.
Basic Flow	1. User (Manager or Supplier) opens the "Track Order" page. 2. System shows a table list of all active orders and their current status. 3. User clicks on a specific order number. 4. System displays the detailed status update timeline with dates and times.
Alternate Flow	None
Exception Flow	Order Number Not Found: 1. User searches for a wrong or invalid order number. 2. System displays a warning: "Order number not found."
Constraints	The status page must load updates quickly from the database within 2 seconds.

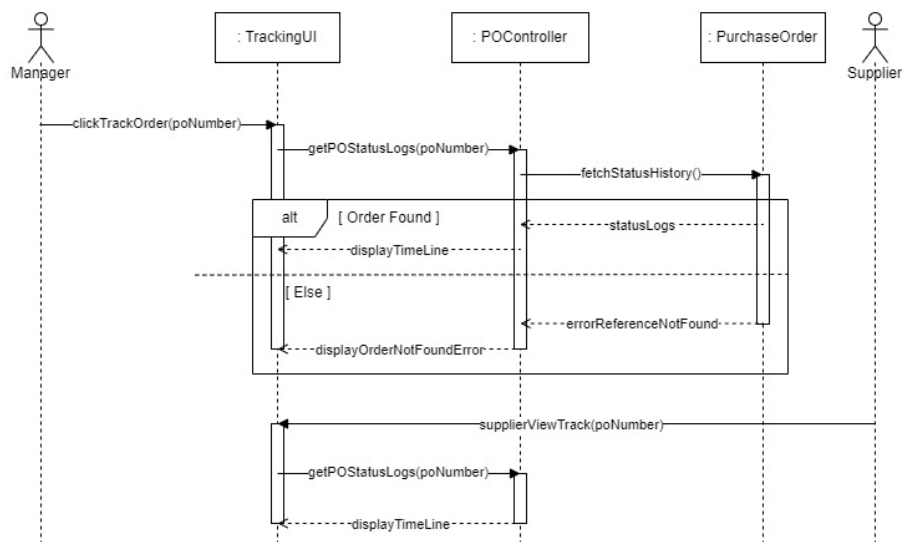
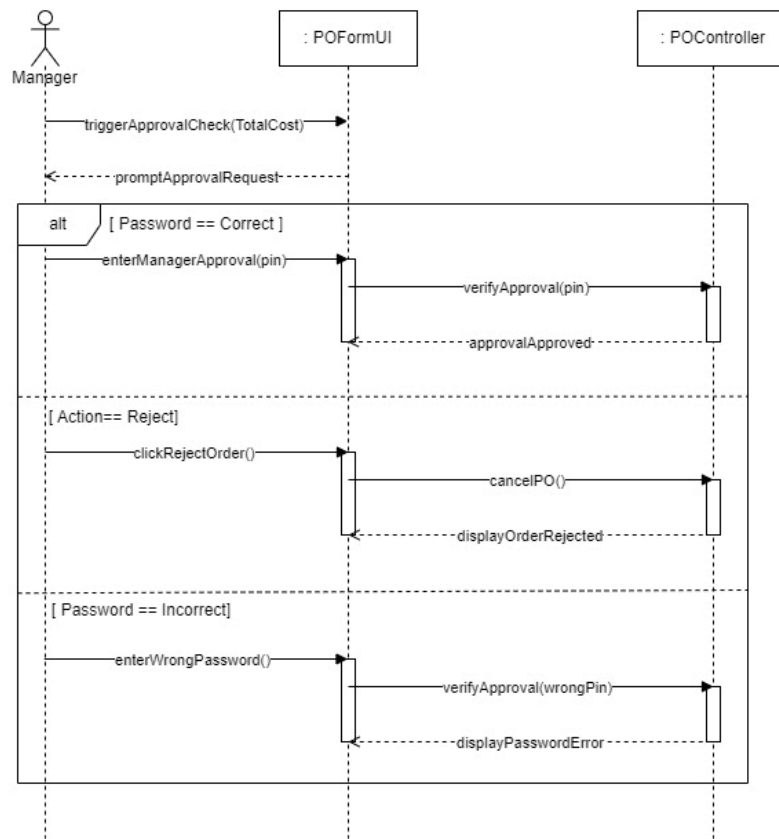
Field	Description
Use Case Name	View Supplier History
Primary Actor	Manager
Secondary Actor	None
Brief Description	Allows the manager to review past orders, old invoices, and performance records of a supplier.
Pre-condition	The supplier has past order records in the system.
Post-condition	The historical transaction data and reports are displayed.
Basic Flow	1. Manager opens the "Supplier History" page. 2. Manager selects a specific supplier name. 3. System collects and displays all old orders and past payments done with this supplier.

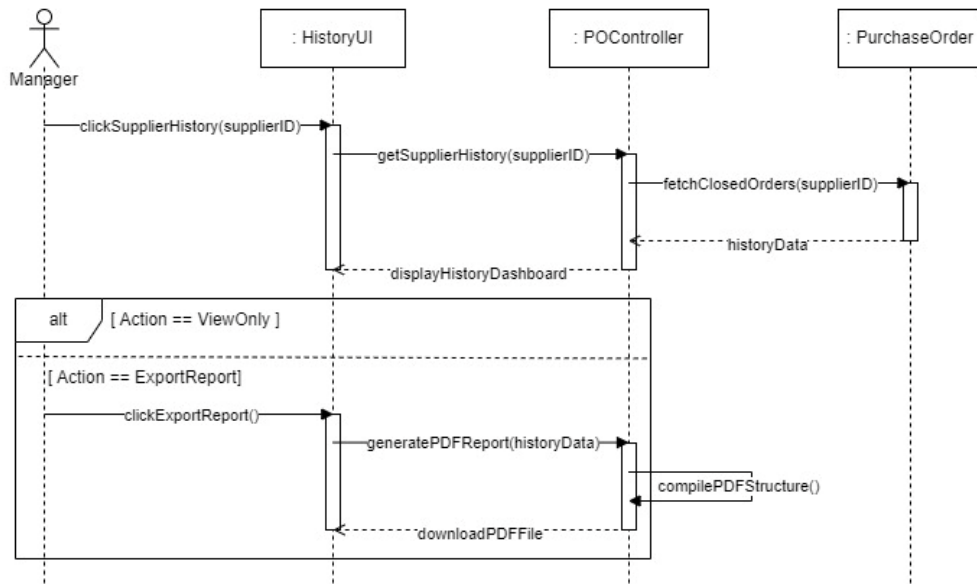
	4. System shows a summary card detailing their average delivery time score.
Alternate Flow	Export Report: 1. Manager clicks the "Download PDF" button. 2. System generates and downloads a summary report file.
Exception Flow	None
Constraints	The system must keep history records up to a maximum rolling period of 5 years.

2.3 SEQUENCE DIAGRAM



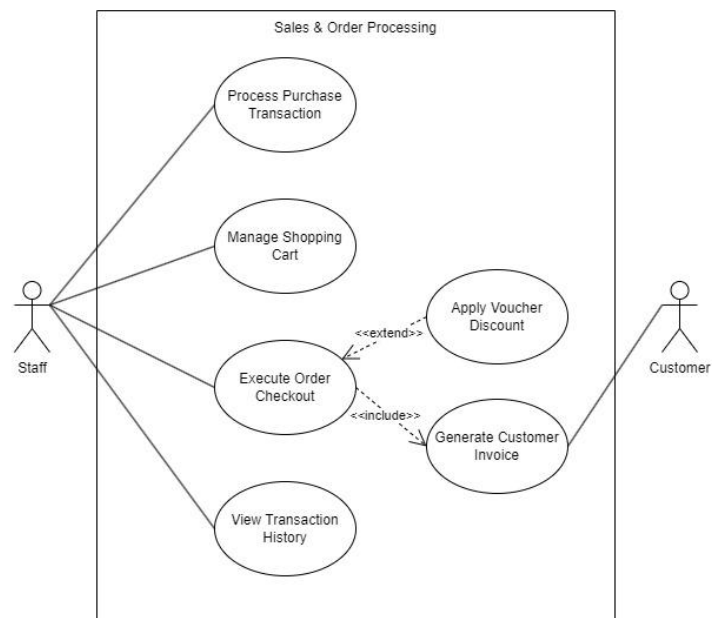






3.0 SALES & ORDER PROCESSING

3.1 USE CASE DIAGRAM



3.2 USE CASE DESCRIPTION

Field	Description
Use Case Name	Process Purchase Transaction
Primary Actor	Staff
Secondary Actor	Customer
Brief Description	Allows the staff to record and process a live purchase transaction when a customer wants to buy items at the counter.
Pre-condition	The Staff is logged into the sales interface.
Post-condition	The transaction details are created and ready for the checkout stage.
Basic Flow	1. Staff clicks the "New Transaction" button. 2. System opens a blank sales entry form. 3. Staff scans or enters the item barcode brought by the customer. 4. System adds the item details to the active screen list.
Alternate Flow	None
Exception Flow	Item Barcode Not Found: 1. System cannot find the scanned barcode in the database. 2. System shows an error: "Item not found. Please enter code manually."
Constraints	The system must load item pricing data from inventory records instantly without lagging.

Field	Description
Use Case Name	Manage Shopping Cart
Primary Actor	Staff
Secondary Actor	None
Brief Description	Allows the staff to add items, update quantities, or remove items from the customer's active shopping basket grid.
Pre-condition	A purchase transaction has already been initialized (UC-01 is active).
Post-condition	The shopping cart items list and item subtotal cost are updated.

Basic Flow	1. Staff views the active item basket list on screen. 2. Staff updates the item quantity (e.g., changes from 1 to 5) or clicks the "Remove Item" icon. 3. System updates the cart items immediately and recalculates the bill subtotal.
Alternate Flow	Clear Whole Cart: 1. Staff clicks the "Clear Cart" button to cancel the entire transaction. 2. System empties the shopping cart and returns to a blank interface.
Exception Flow	None
Constraints	Cart item quantity fields must only accept positive integers greater than zero.

Field	Description
Use Case Name	Execute Order Checkout
Primary Actor	Staff
Secondary Actor	None
Brief Description	Handles the final checkout step where the staff processes the customer payment and finalizes the transaction.
Pre-condition	The shopping cart contains at least one valid item.
Post-condition	The customer payment is confirmed, stock levels are deducted, and status becomes "Completed".
Basic Flow	1. Staff clicks the "Proceed to Checkout" button. 2. System shows the payment selection screen. 3. Staff enters the cash amount received from the customer. 4. System automatically calculates the final bill price and change due. 5. Staff clicks "Confirm Payment". 6. System executes the mandatory Generate Customer Invoice function.
Alternate Flow	None
Exception Flow	Insufficient Payment Amount: 1. Staff enters a payment value lower than the order total cost.

	2. System blocks confirmation and shows a warning: "Payment amount insufficient."
Constraints	External payment gateway transactions must receive status verification before saving records.

Field	Description
Use Case Name	Apply Voucher Discount
Primary Actor	Staff
Secondary Actor	None
Brief Description	Deducts a percentage or flat cash discount from the total bill if the customer provides a promo voucher code.
Pre-condition	The order checkout phase is currently active.
Post-condition	The order grand total is reduced based on the active voucher value constraints.
Basic Flow	1. Staff clicks the "Apply Promo/Voucher" button option. 2. Staff types the customer's voucher alphanumeric code and clicks "Verify". 3. System validates the voucher rules, calculates the discount value, and minuses it from the grand total.
Alternate Flow	None
Exception Flow	Voucher Code Expired/Invalid: 1. System checks the database logs and finds the code is invalid or past its expiry date. 2. System displays an error notice banner: "Voucher code invalid or expired."
Constraints	The system must prevent stacking multiple voucher discount codes on a single invoice transaction bill.

Field	Description
Use Case Name	Generate Customer Invoice
Primary Actor	Staff
Secondary Actor	Customer
Brief Description	System automatically compiles checkout metrics to generate and print a formal sale invoice document for the customer.
Pre-condition	Payment authorization completes successfully within Execute Order Checkout.
Post-condition	A final digital invoice record is archived, and the paper document is printed out for the Customer.
Basic Flow	1. System aggregates transaction metadata (Timestamp, Itemized grid list, Tax tiers, Final amount paid). 2. System commits the record and provisions a unique Invoice ID tracking code row string. 3. System routes the document output commands to print the physical invoice layout for the Customer.
Alternate Flow	None
Exception Flow	None
Constraints	Generated digital invoice parameters must reflect final payment balances accurately down to the cent.

Field	Description
Use Case Name	View Transaction History
Primary Actor	Staff
Secondary Actor	None
Brief Description	Allows the staff to view, look up, and inspect historical customer invoices and completed sales logs.
Pre-condition	Historical sales data rows exist inside long-term server memory stores.
Post-condition	Past transaction row records load across the system screen query viewport layouts.
Basic Flow	1. Staff clicks the "Sales History Logs" sidebar dashboard link option. 2. System renders an indexing log grid summarizing past sales

	metrics. 3. Staff inputs an Invoice ID serial code token string into the lookup search bar utility. 4. System filters results and presents the matching invoice breakdown view.
Alternate Flow	None
Exception Flow	Invoice ID Serial Not Found: 1. Staff types a non-existent or wrong invoice reference serial ID string. 2. System clears display grids and throws an alert message: "Transaction record reference not found."
Constraints	System logging dashboards must securely populate text index rows within 2 seconds.

3.3 SEQUENCE DIAGRAM

